MATEC Web of Conferences **287**, 02003 (2019) https://doi.org/10.1051/matecconf/201928702003
Power Transmissions 2019

Durability Investigation of plastic gears

Gorazd Hlebanja[1,*], *Matija* Hriberšek[2], *Miha* Erjavec[2], and *Simon* Kulovec[2]

[1]Podkrižnik d.o.o., Lesarska c. 10, 3331 Nazarje; University of Novo mesto, Na Loko 2, 8000 Novo mesto, Slovenia
[2]Podkrižnik d.o.o., Lesarska c. 10, 3331 Nazarje, Slovenia

Abstract. Plastic materials can be used for various gears and gear mechanisms due to their qualities, like wear resistance when running dry, low noise, vibration damping, low inertia, low manufacturing cost, etc. However, they also expose with temperature connected problems like rapidly decreasing load, thermal expansion and accuracy of moulded gears. Therefore, it is a challenge to produce gears with adequate accuracy and durability to serve in contemporary mechatronic applications in automotive, in medical equipment and many other industries. Such materials are POM, PA6, PA66, PEEK, reinforced plastic materials, etc. and steel. Proper combination of such materials can result in long and reliable operation of such mechanisms. On the other hand, S-gears offer improved properties in comparison to involute (E-gears), which essentially comprise lower contact pressure, less sliding and frictional losses, stronger root and improved oil film when lubricated. The goal is development of advanced power transmission drive components made of polymers. So, a material database, a testing system and testing procedures with varying material combinations and gear pairs according to VDI are necessary to support a designer in proper decisions. Some material properties, especially fillers are presented. Furthermore, testing facilities, gears and some results are discussed.

1 Introduction

One can agree on advantages of plastic gearings, which are low density resulting in reduced weight and lower moment of inertia, elasticity and impact resistance, low coefficient of friction, corrosion resistance, manufacturing cost in terms of mass production, good performance in noise, vibration and harshness. Limitations of plastic gears are in lower mechanical strength which result in lesser load-carrying capacity compared to metal gears. On the other hand, the number of available materials is growing almost at an exponential rate and various companies produce materials with similar properties as well. Proper combination of such materials can result in long and reliable operation of such mechanisms and a cost-effective solution.

The paper presents some actual material combinations for gears, which can be used in complex parts. Due to immense development in materials it is necessary to assess their properties, their design value. The simplest way is to experimentally estimate durability of such material with simple, yet accurate testing rigs. Results can be then compared to the other available materials and an optimal material can be used in the design.

Since the VDI Richtlinie Blatt 4 [1] proposes standardised testing procedure for mould injected and cut gears it was necessary to design and implement a new testing rig to accommodate conforming tests.

Some tests with small plastic gears conducted to assess Wöhler curves for gears with various material combinations and long-term tests with smaller loads to run up to 10 or 20 million cycles in the controlled environment have been conducted recently [2-5]. Results comparing S- and E-gear shape indicate better performance of S-gears. S-gear geometry feature lower contact pressure, less sliding and frictional losses, stronger root and improved oil film when lubricated.

Thermal models should reflect actual gear tooth shape, which is an unaccomplished task, yet. So, frictional power along the path of contact, work of friction and deriving flash temperatures along the path of contact calculated for both gear types disclose lower thermal load of S-gears [3].

The main goal is development of advanced power transmission drive components from polymeric materials (gear pairs), which will suit the customer demands and improve gearbox lifetime. It includes optimization and development of materials, improving guidelines for gear and tool designing and expanding of material database in terms of Wohler's curves, wear coefficients, temperatures in meshing zone for typical polymeric materials, which will be the main contribution for design engineer in a real industrial environment.

2 Thermoplastic materials for gears

Mechanical properties of basic plastic materials can be